Router:-
A router lets you navigate between different pages in a React app without refreshing the browser.
Instead of loading a new HTML file like traditional websites, React swaps components dynamically.

Without routing:

- You only have one page
- Everything loads in the same component

With routing:

- / → Home page
- /about → About page
- /contact → Contact page
- /dashboard → Dashboard page

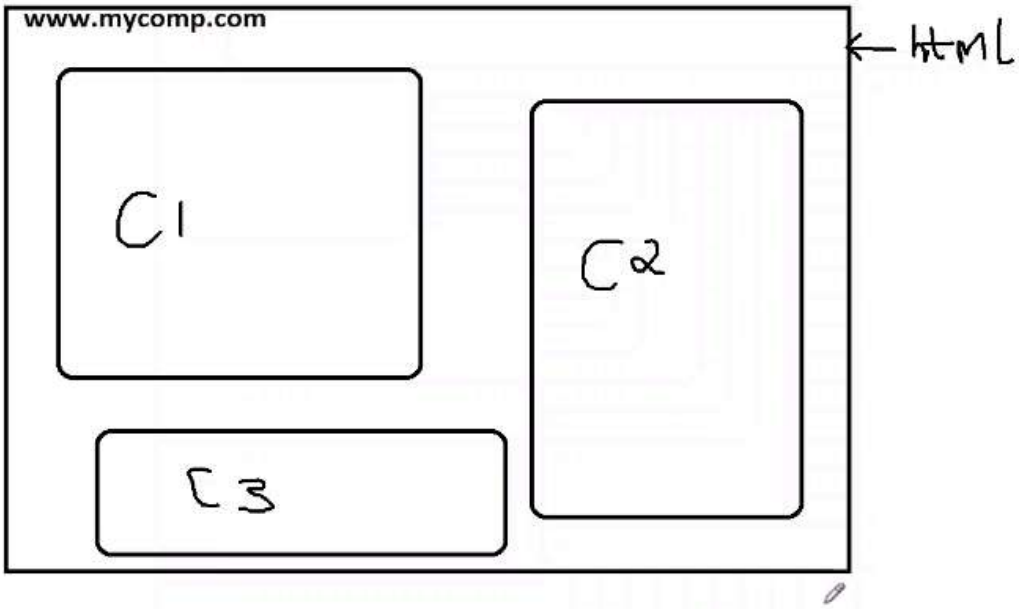
And React changes the component instantly

First we have to install router:-

npm install react-router-dom

Real life example:-

- Hospital Management System
- IT Ticket Triage System
- Instagram Analyzer



Create a complete React E-commerce project and zip it

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="utf-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1" />
  <title>Flipkart Mini</title>
</head>
<body>
  <div id="root"></div>
</body>
</html>
```

index.js

```
import React from "react";
import ReactDOM from "react-dom/client";
import App from "./App";

const root = ReactDOM.createRoot(document.getElementById("root"));
root.render(<App />);
```

App.js

```
import React, { useState } from "react";
import { BrowserRouter, Routes, Route } from "react-router-dom";
import Navbar from "./components/Navbar";
import Loginform from "./components/Loginform";
import Product from "./components/Product";
import Shopping from "./components/Shopping";

function App() {
  const products = [
    { id: 1, title: "iPhone 15", price: 79999 },
    { id: 2, title: "Samsung S24", price: 69999 },
    { id: 3, title: "Boat Headphones", price: 1999 }
  ];

  const [cart, setCart] = useState([]);

  const addToCart = (product) => {
    setCart((prev) => {
      const exist = prev.find((p) => p.id === product.id);
      if (exist) {
        return prev.map((p) =>
          p.id === product.id ? { ...p, qty: p.qty + 1 } : p
        );
      }
      return [...prev, { ...product, qty: 1 }];
    });
  };

  const cartCount = cart.reduce((sum, item) => sum + item.qty, 0);

  return (
    <BrowserRouter>
      <Navbar cartCount={cartCount} />
      <Routes>
        <Route path="/" element={<Product products={products} addToCart={addToCart} />} />
        <Route path="/cart" element={<Shopping cart={cart} />} />
        <Route path="/login" element={<Loginform />} />
      </Routes>
    </BrowserRouter>
  );
}

export default App;
```

navbar.js

```
import React from "react";
import { Link } from "react-router-dom";

function Navbar({ cartCount }) {
  return (
    <div style={{ padding: 15, background: "#0d6efd", color: "white", display: "flex", justifyContent:
"space-between" }}>
      <h2>Flipkart Mini</h2>
      <div>
        <Link to="/" style={{ margin: 10, color: "white" }}>Products</Link>
        <Link to="/cart" style={{ margin: 10, color: "white" }}>Cart ({cartCount})</Link>
        <Link to="/login" style={{ margin: 10, color: "white" }}>Login</Link>
      </div>
    </div>
  );
}

export default Navbar;
```

product.js

```
import React from "react";

function Product({ products, addToCart }) {
  return (
    <div style={{ padding: 20 }}>
      <h2>Products</h2>
      {products.map((p) => (
        <div key={p.id} style={{ border: "1px solid #ccc", padding: 10, margin: 10 }}>
          <h3>{p.title}</h3>
          <p>₹ {p.price}</p>
          <button onClick={() => addToCart(p)}>Add to Cart</button>
        </div>
      ))}
    </div>
  );
}

export default Product;
```

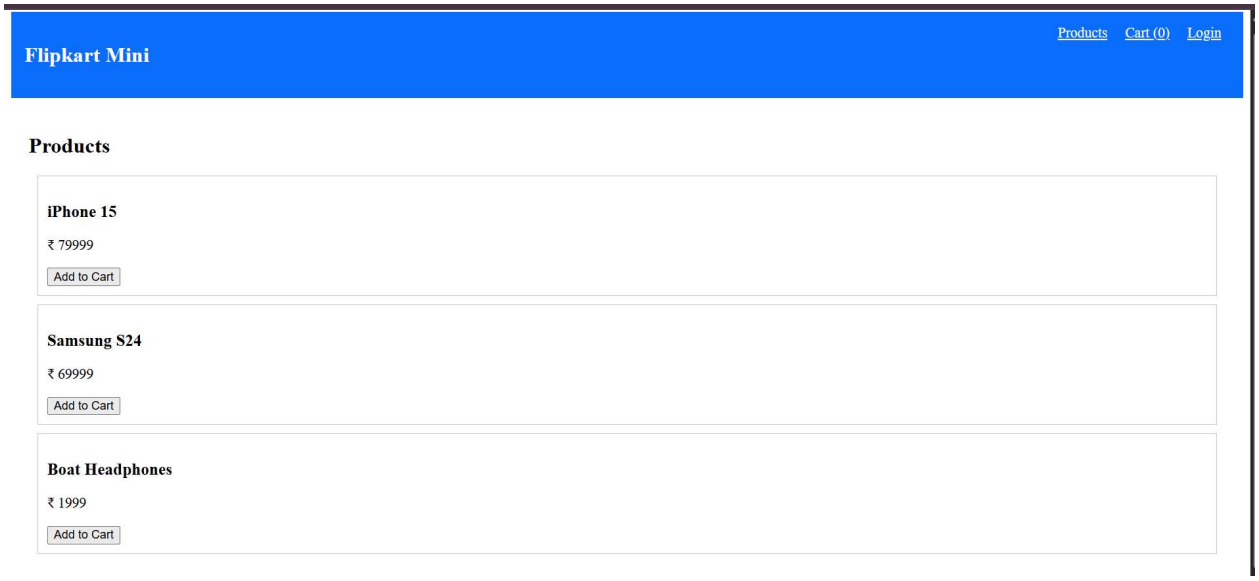
shopping.js

```
import React from "react";

function Shopping({ cart }) {
```

```
  const total = cart.reduce((sum, item) => sum + item.price * item.qty, 0);

  return (
    <div style={{ padding: 20 }}>
      <h2>Cart</h2>
      {cart.length === 0 ? (
        <p>Cart is empty</p>
      ) : (
        <>
          {cart.map((item) => (
            <div key={item.id}>
              {item.title} - {item.qty} x ₹{item.price}
            </div>
          ))}
          <h3>Total: ₹ {total}</h3>
        </>
      )}
    </div>
  );
}

export default Shopping;
```

login.js

```
import React from "react";

function Loginform() {
  return (
    <div style={{ padding: 20 }}>
      <h2>Login Page</h2>
      <input placeholder="Email" /><br /><br />
      <input type="password" placeholder="Password" /><br /><br />
      <button>Login</button>
    </div>
  );
}

export default Loginform;
```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